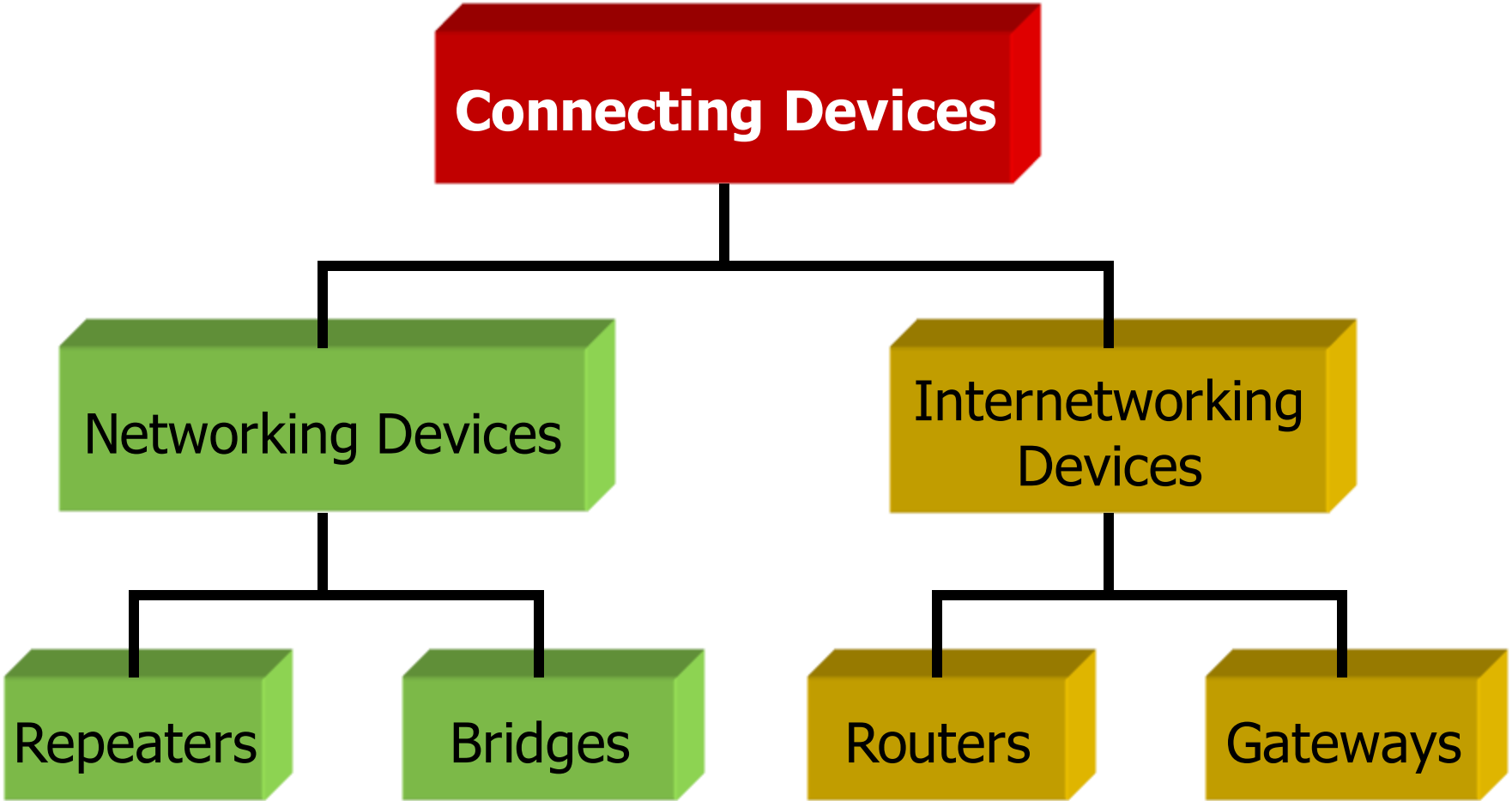


Connecting Devices



```
graph TD; A[Connecting Devices] --> B[Networking Devices]; A --> C[Internetworking Devices]; B --> D[Repeaters]; B --> E[Bridges]; C --> F[Routers]; C --> G[Gateways];
```

Networking Devices

Internetworking Devices

Repeaters

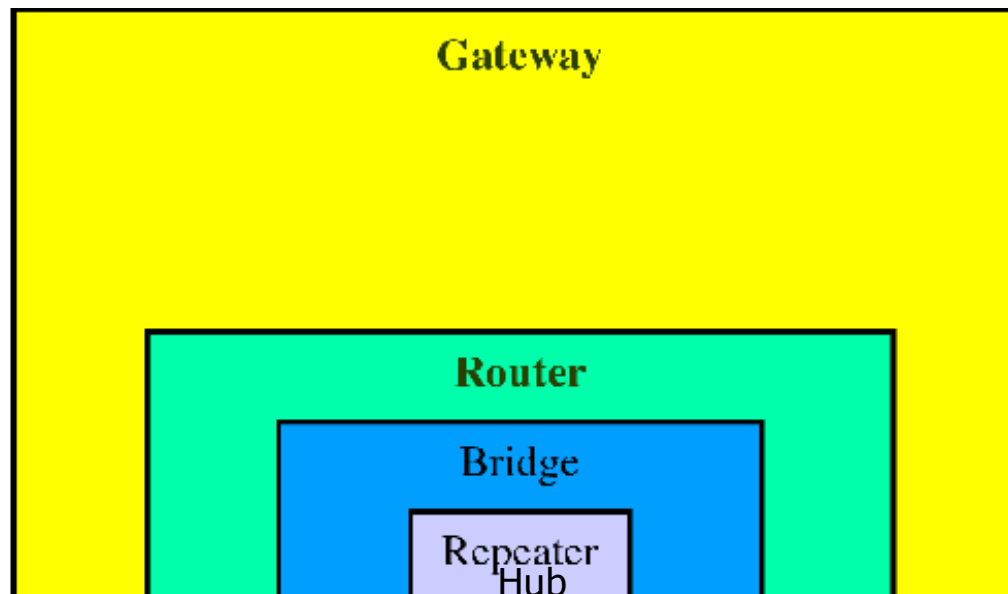
Bridges

Routers

Gateways

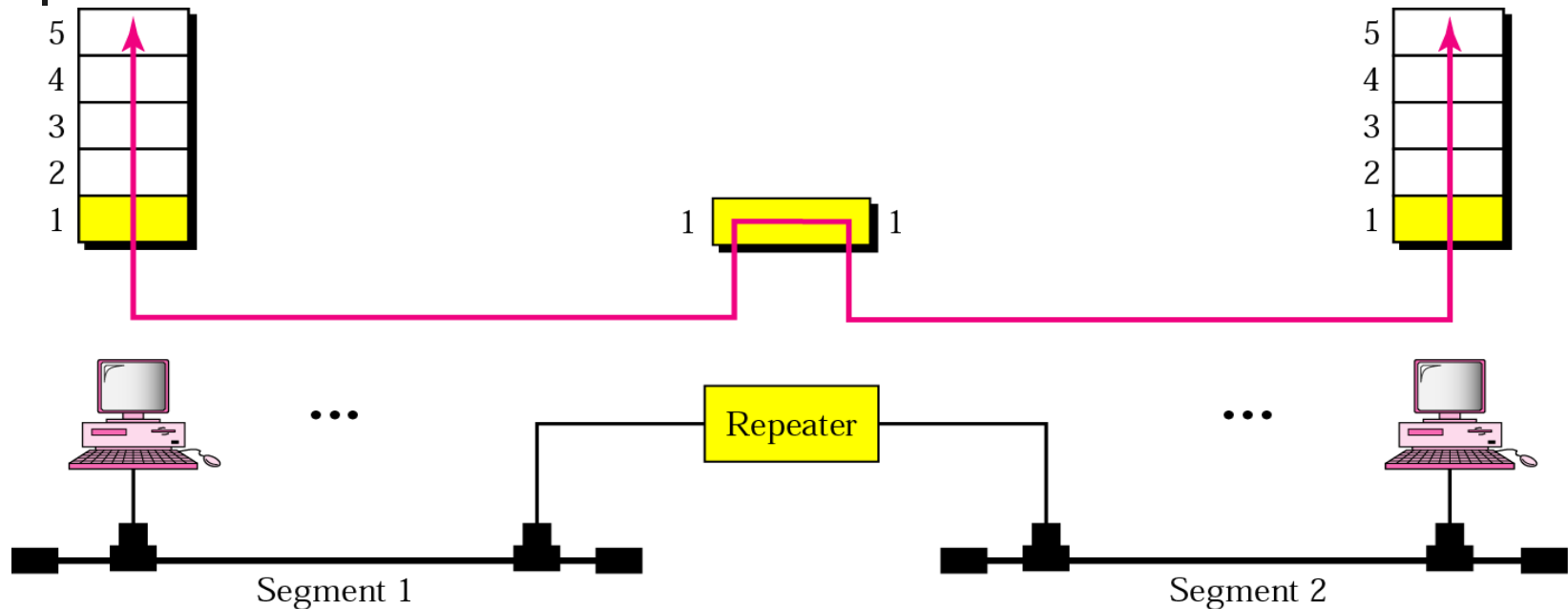
Connecting Devices

Application
Presentation
Session
Transport
Network
Data link
Physical



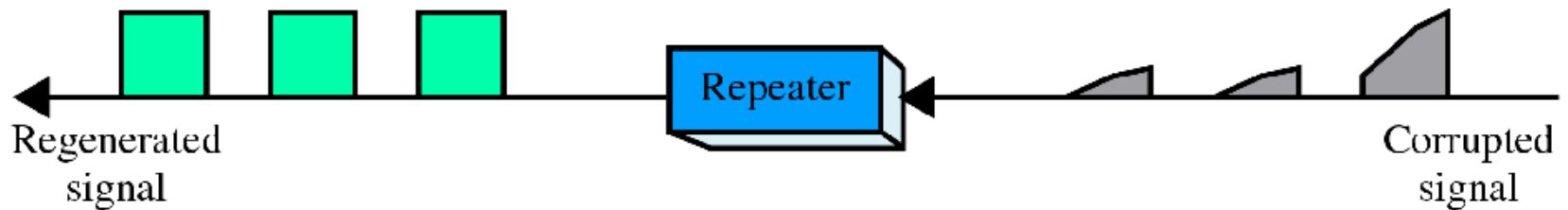
Application
Presentation
Session
Transport
Network
Data link
Physical

Repeater

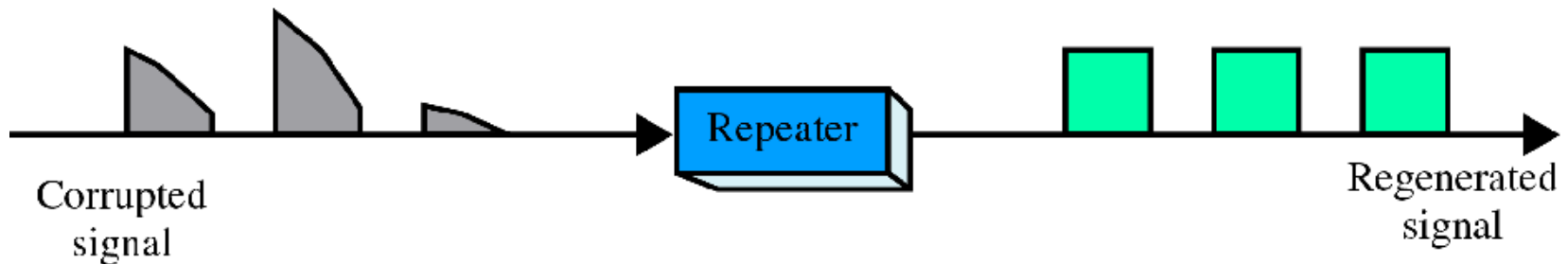


- Extend the physical length
- No network function has been changed
- Location is matter

Function of repeater



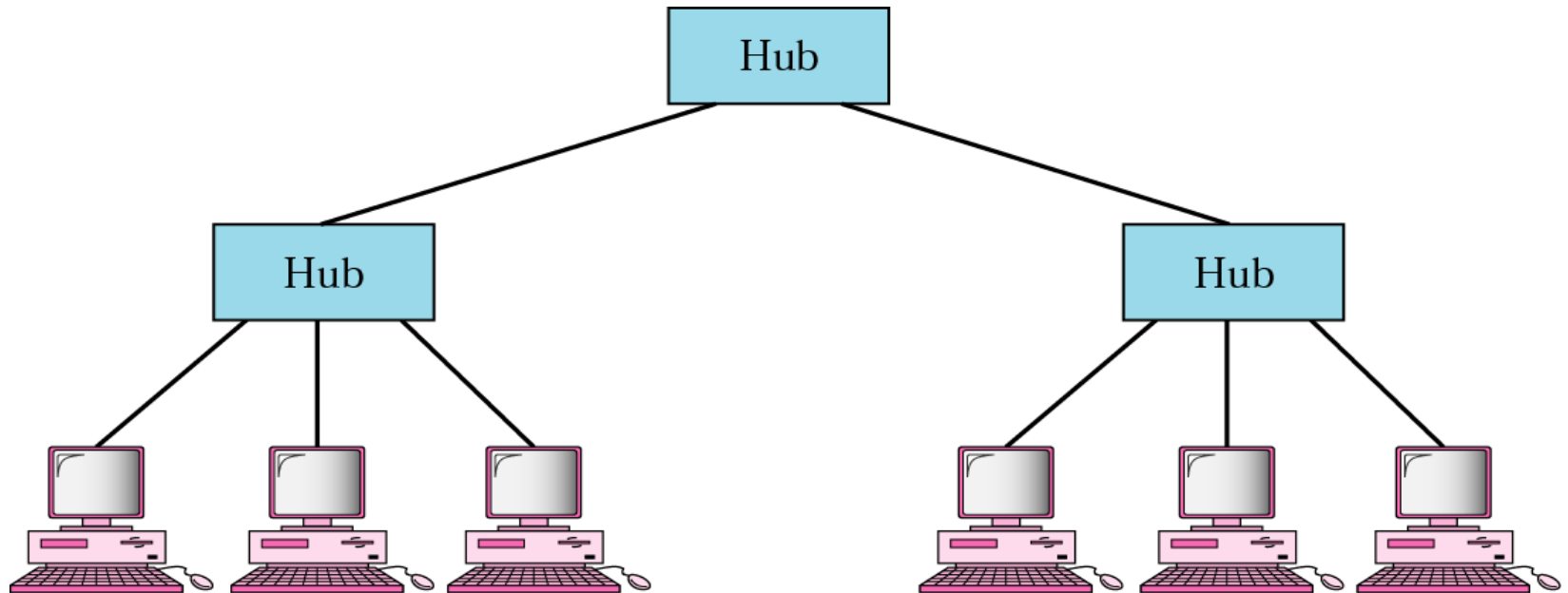
(a) Right-to-left transmission.



(b) Left-to-right transmission.

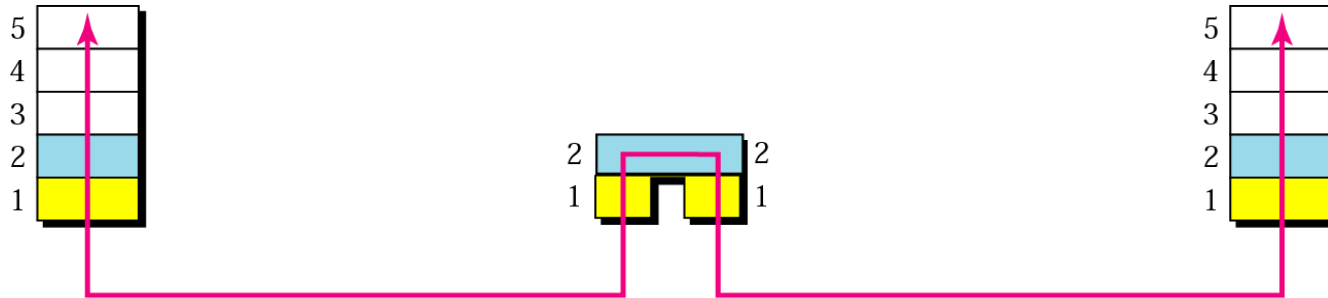
Repeater is not exactly as same as Amplifier

Hub



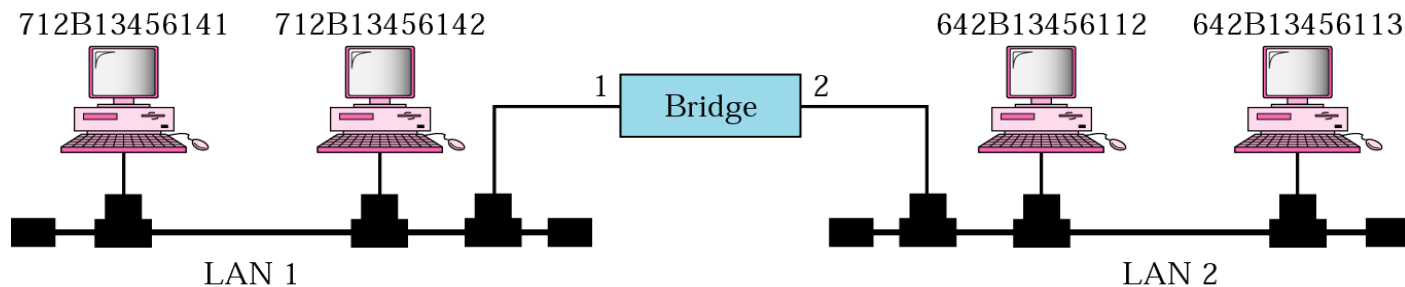
- Actually is a multiport repeater
- Star / Tree Topology

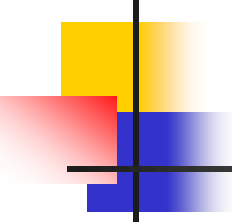
Bridge



Address	Port
712B13456141	1
712B13456142	1
642B13456112	2
642B13456113	2

Bridge Table

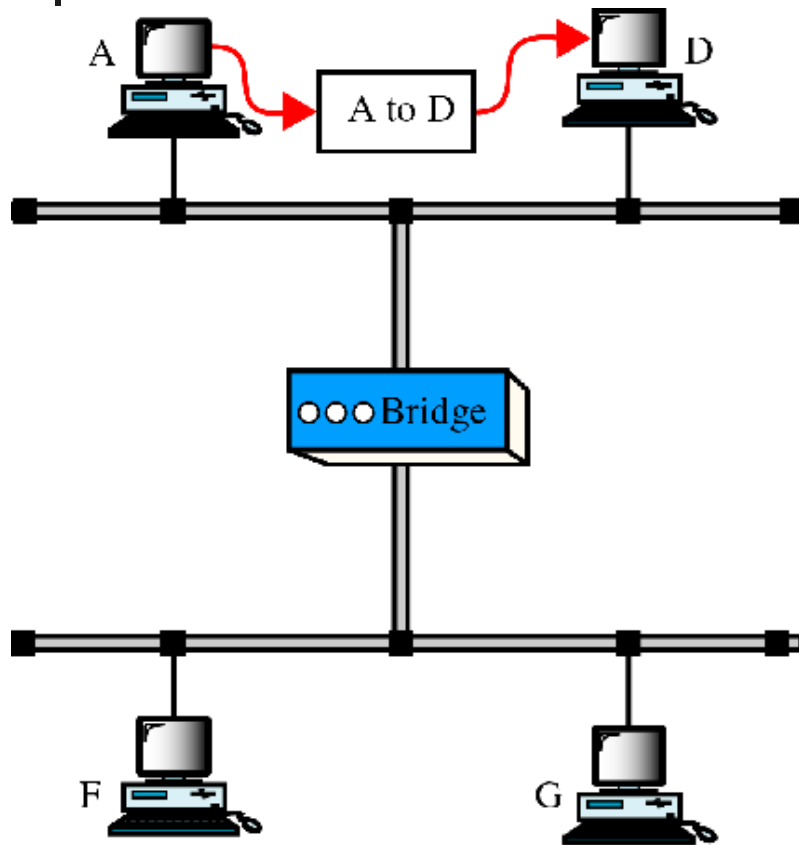




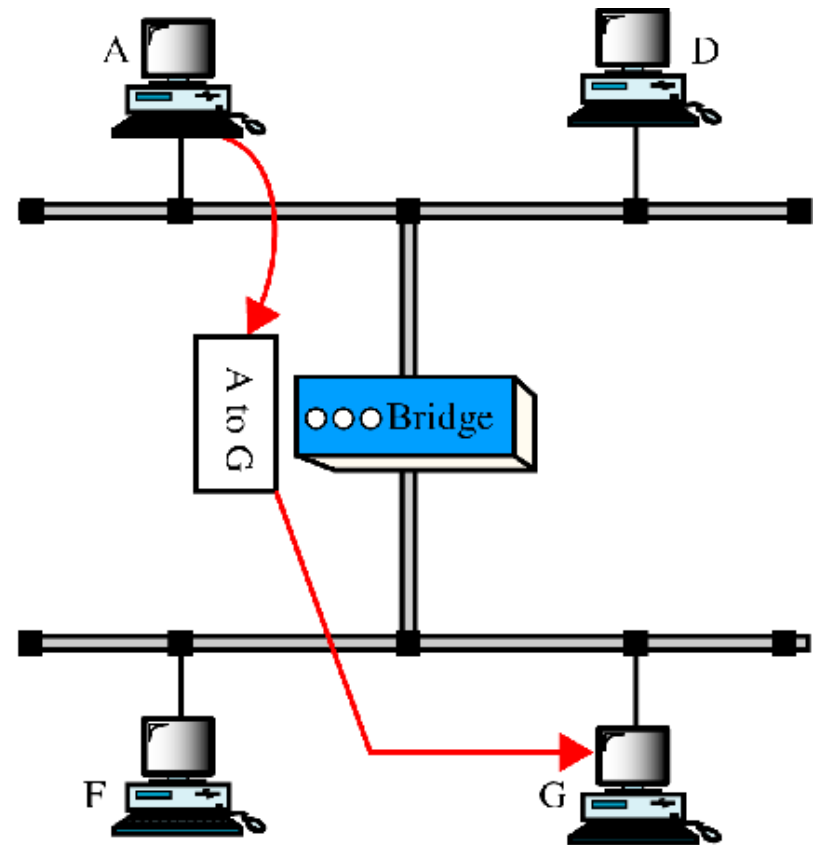
Bridge

- Divide a large network into smaller segment
- Isolating and controlling the link problems (e.g. congestion)
- Regenerate signal + Checking Physical Address and forward only to the specified segment

Function of a bridge

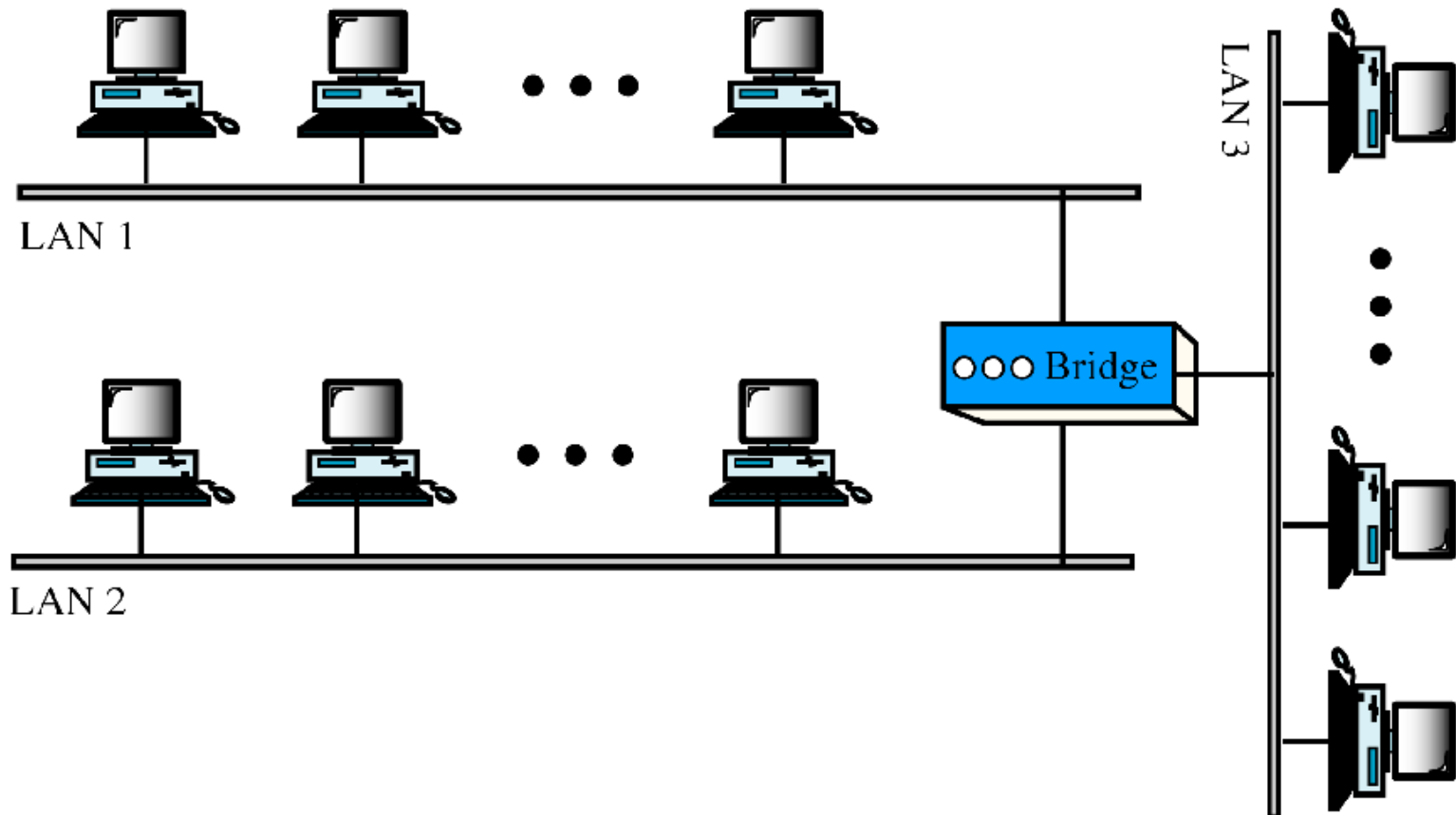


a. A packet from A to D

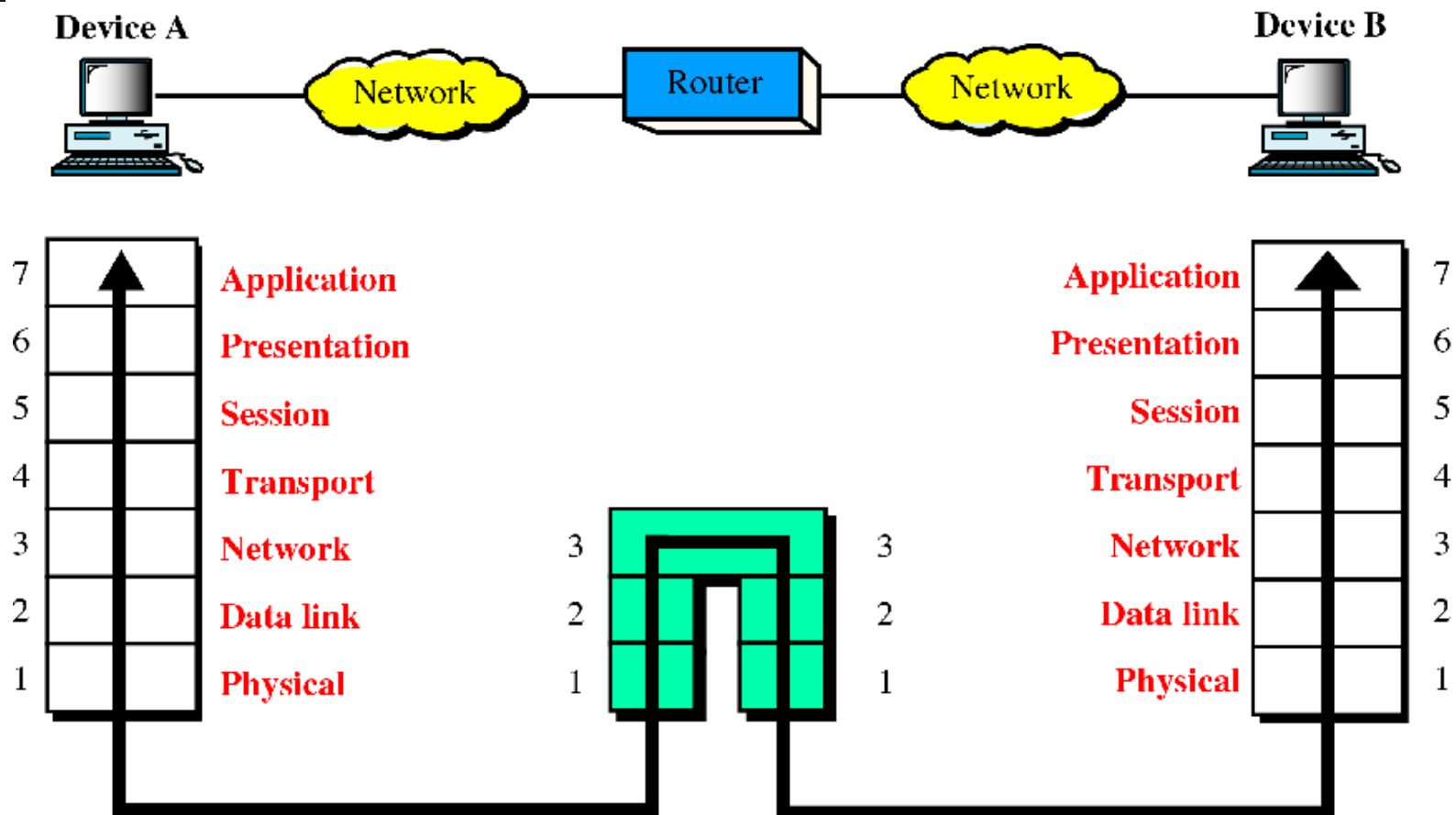


b. A packet from A to G

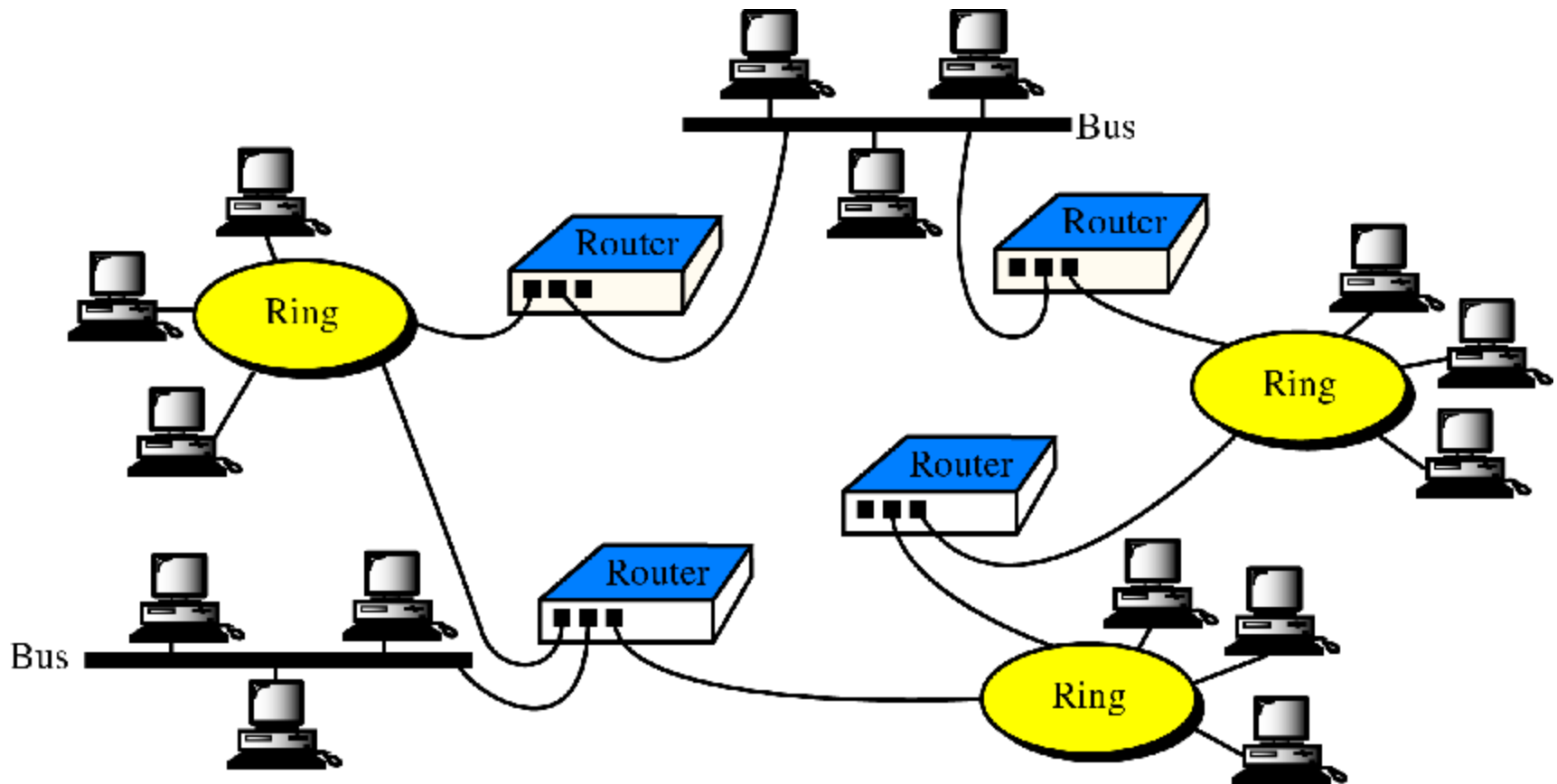
Multiport bridge

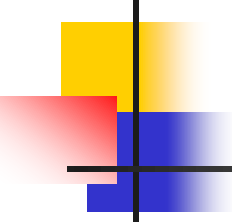


Routers



Routers in an internet



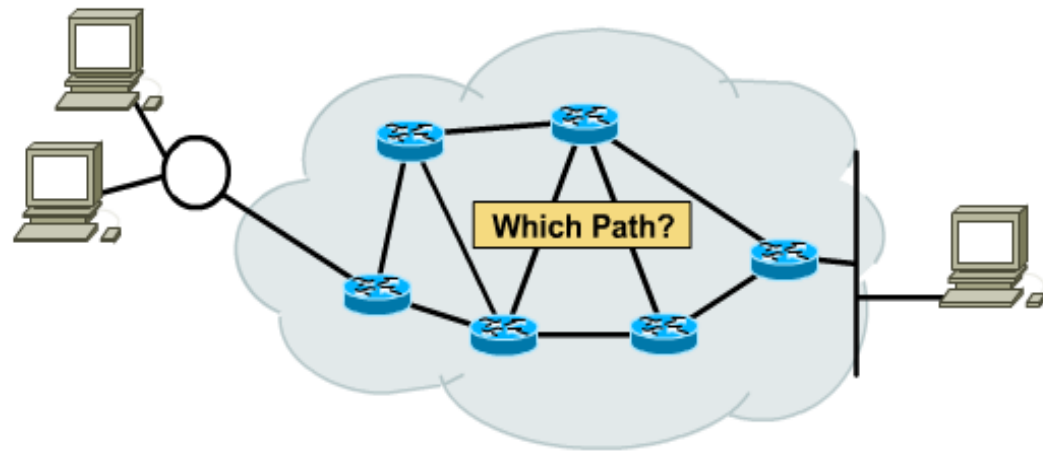


Routers

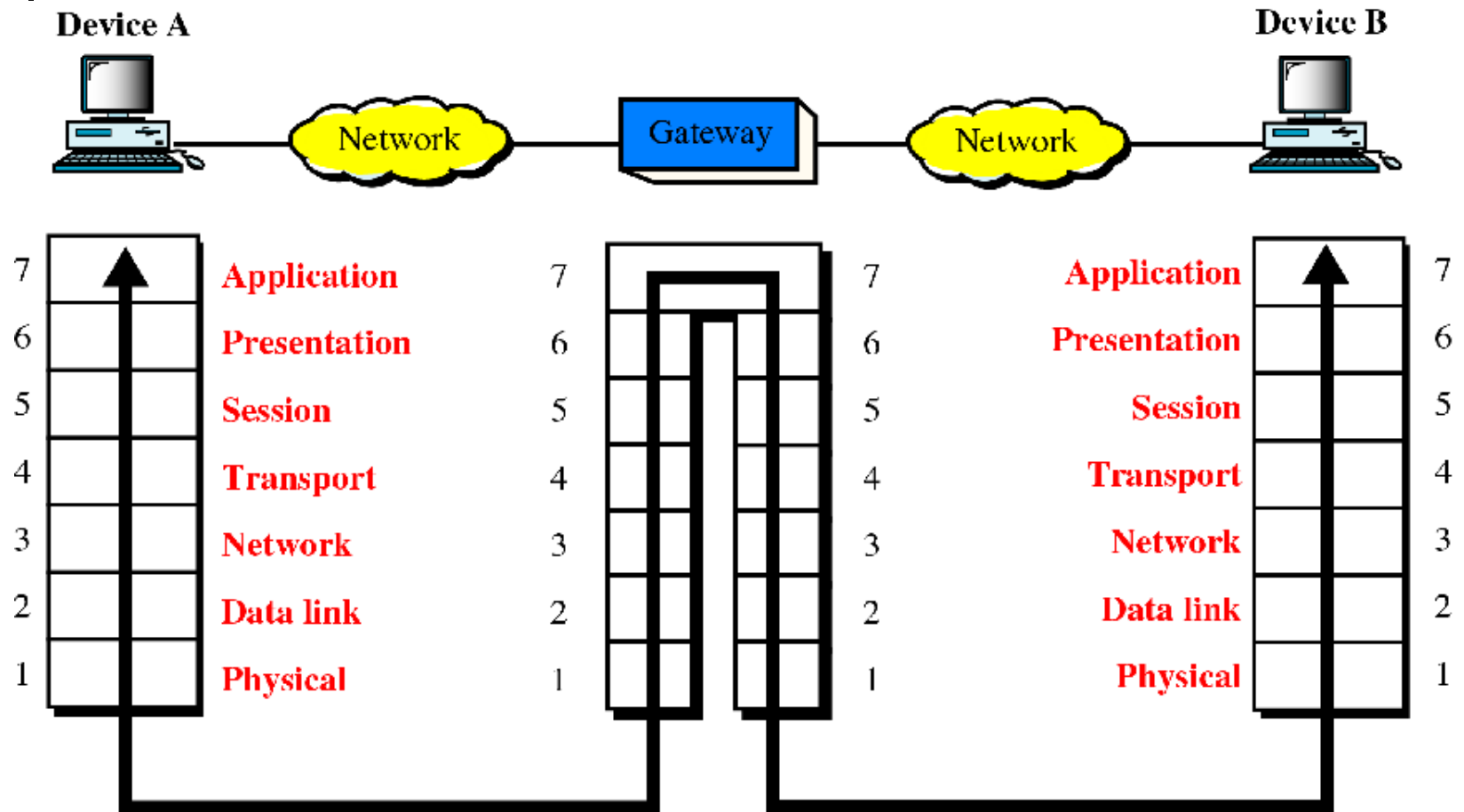
- Act like stations on a network
- Multi-home
- Definition (Goal)
 - “Learning how to get from here to there.”
 - “Process of discovering, selecting, and employing paths from one place to another (or to many others) in a network” [from David M. Piscitello, Bellcore and A. Lyman Chapin, BBN]

Routing Principle

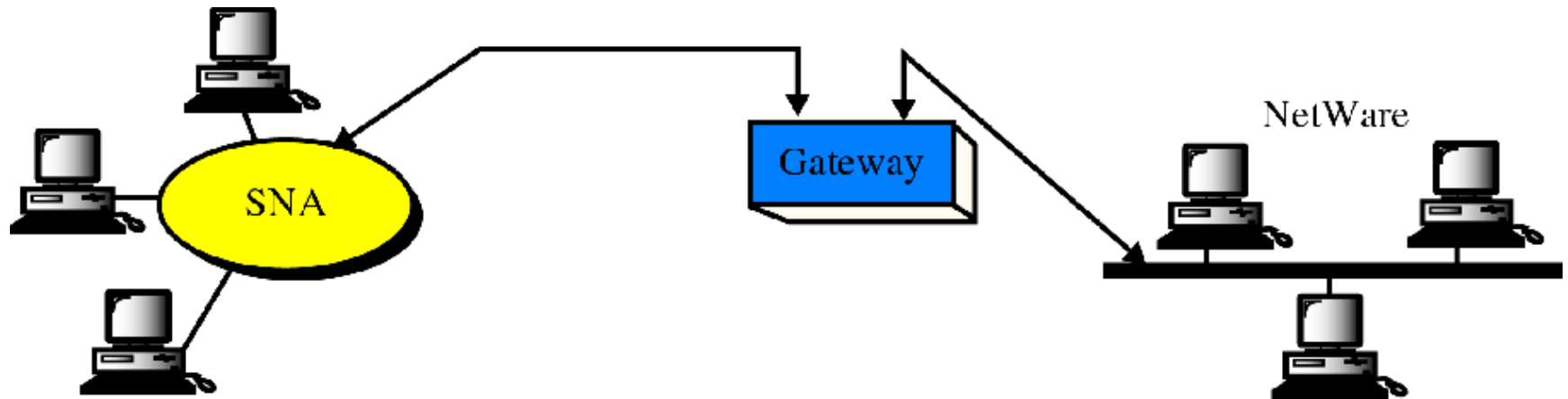
- Goal: Arriving at the destination
- Considerations:
 - Direct route (shortest)
 - Reliable route
 - Cheap route
 - Safe route
 - Scenic route ☺



Gateways (protocol converter)



A gateway



SNA network (IBM)
Netware network (Novell)